INCH POUND

A-A-52550B
September 22, 2026
SUPERSEDING
A-A-52550A
June 30, 2023

COMMERCIAL ITEM DESCRIPTION

PINTLE ASSEMBLY, TOWING, MANUAL RELEASE, 18 000, 40 000 AND 100 000 LBS CAPACITY

The General Services Administration has authorized the use of this Commercial Item Description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers manual release pintle assemblies that are used on trucks and equipment designed for over the road towing within the specified capacities.

2. CLASSIFICATION. Pintle assemblies covered by this CID are listed by type and maximum gross trailer weight (GTW) as follows:

Type I - 18 000 pound (lb) maximum (max.) GTW
Class 1 - Swivel
Class 2 - Rigid
Type II - Swivel 40 000 lb max. GTW
Type III - Swivel 100 000 lb max. GTW

3. SALIENT CHARACTERISTICS

3.1 Materials. Unless otherwise specified herein, materials shall be in accordance with the manufacturer's materials specifications for pintle assemblies. The use of recovered materials made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see paragraph 4.1).

3.1.1 Hexavalent chromium and cadmium free finishes. Pintle assemblies with an AA52550-*E PIN (see paragraph 7.3) shall be free of hexavalent chromium and cadmium containing finishes.

Note: * in 3.1.1 indicates placeholder since there are four possibilities (1, 2, 3 or 4).

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FSC 2540

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3.2 Design and construction. Unless otherwise specified herein, the design and construction of the pintle assemblies shall be in accordance with figures 1 through 4 for the applicable type required.

3.2.1 Secondary locking mechanism. The secondary locking mechanism shall consist of either a cotter pin with an attached chain or an equivalent locking device such as a snapper pin and lanyard or a spring-loaded pin (see paragraph 7.4). The locking device shall not require the use of any special tools to open or close it. The finish on the locking device and attachment chain or lanyard shall not require any additional paint or top coat.

3.2.2 Lubrication fitting. Lubrication fitting shall be in accordance with SAE J534 and shall have 0.25-28 taper threads.

3.2.3 Fastener Dimensions. All fasteners shall be dimensioned in accordance with IFI standards where applicable.

3.2.4 STANAG requirements. Pintles shall be compliant with the requirements of STANAG 4101 (see paragraph 7.7) for articulation and acceptance of required drawbar eye.

3.3 Painting. Cleaning, surface treatment, and painting shall be as specified in the contract (see paragraph 7.2). If unspecified, paint with the CARC system in accordance with MIL-DTL-53072, exterior topcoat color Green 383, 34094.

3.4 Identification and marking. Identification and marking of the pintle assemblies shall include, as a minimum, the part identification number (PIN), maximum GTW, and the manufacturer's CAGE code (or equivalent identification) (see paragraph 7.3).

3.5 Load capacity. Maximum GTW shall be in accordance with SAE J847.

3.6 Corrosion resistance. All fasteners and hardware components, exclusive of lubrication fittings (see paragraph 3.2.2), of the pintle shall be either made of corrosion resistant material (CRES, brass, composite, etc.) or plated or coated to prevent corrosion.

3.6.1 Salt spray requirement. For hexavalent chromium and cadmium free parts (E Code, see paragraph 7.3), all platings and coatings shall be certified to meet the following minimum corrosion performance per ASTM B117: No appearance of corrosion products of the protective coating or base metal before 96 hours. Figures 1-4 identify hardware that must meet this requirement. Replaceable parts such as D-clips, cotter pins, spring-loaded pins, chains, U-drive screws, and snapper pins are exempt from this requirement.

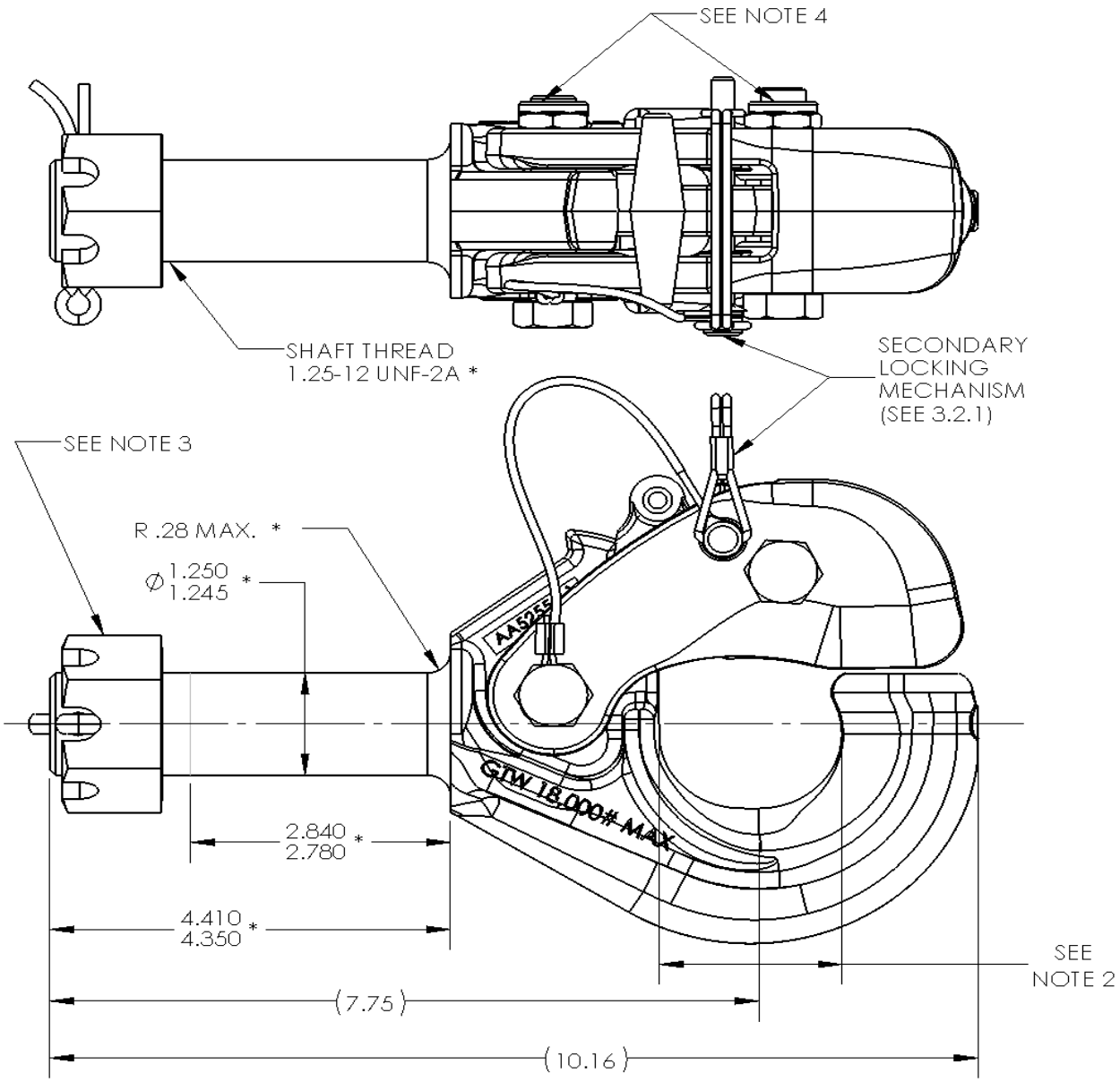
4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.2 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

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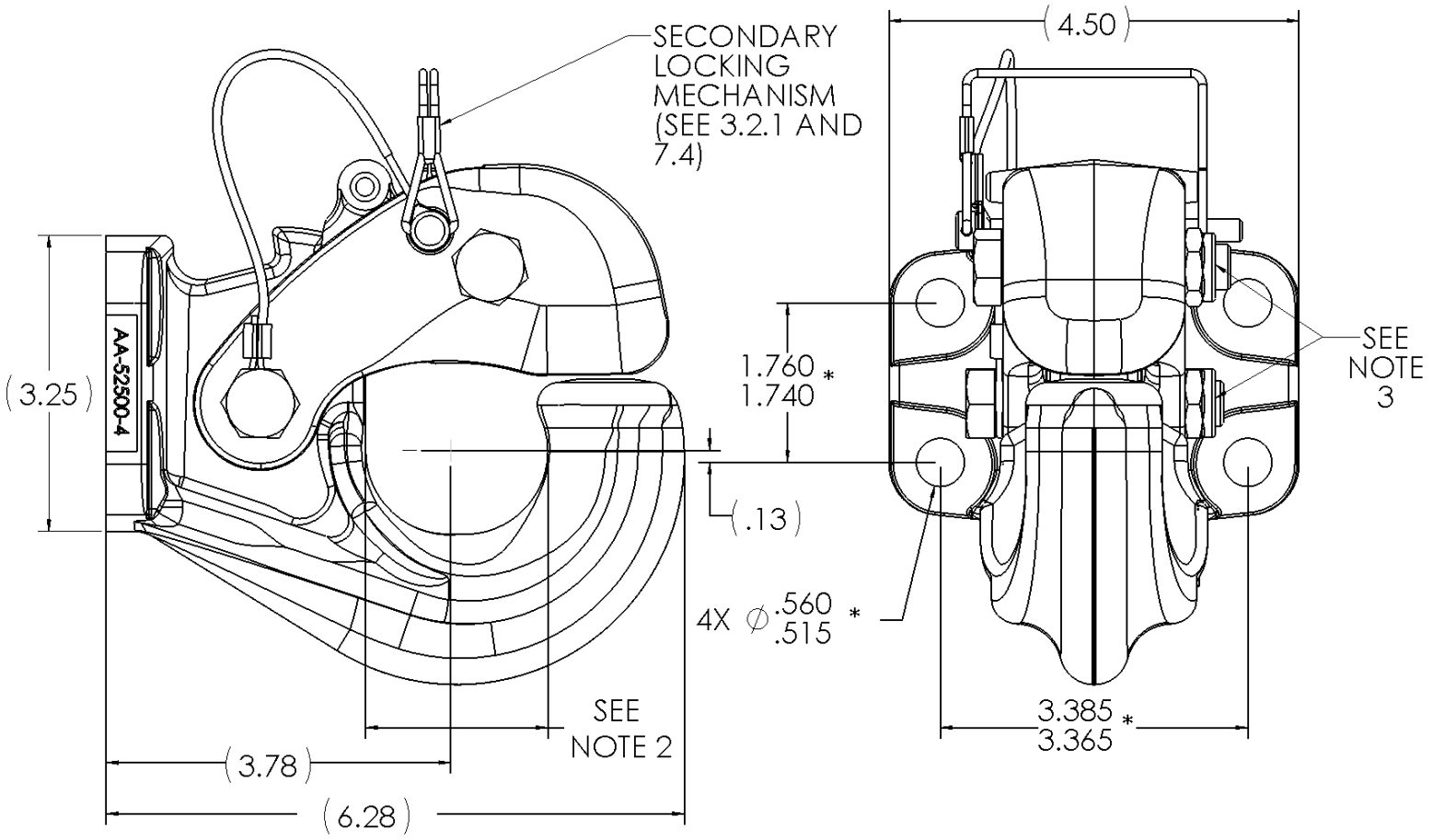
PIN		Former MS Part No.
May contain Cad & Cr6+	Cad & Cr6+ Free	MS51335-1
AA52550-1	AA52550-1E	

NOTES:

1. Dimensions are in inches. *These dimensions are required, all other dimensions are for engineering reference only.
2. The pintle throat shall be sized such that it will accept a 3 in. ID. X 1.66 dia. cross section lunette per STANAG 4101 (see paragraph 7.7).
3. Castellated nut, cotter pin, and shaft thread configuration are required as indicated.
4. Positive or prevailing locking device required to prevent loosening during service
5. Castellated nut and latch mounting hardware (note 3 & 4) to comply with paragraphs 3.6 and 3.6.1.

FIGURE 1. Type I, class 1, pintle assembly.

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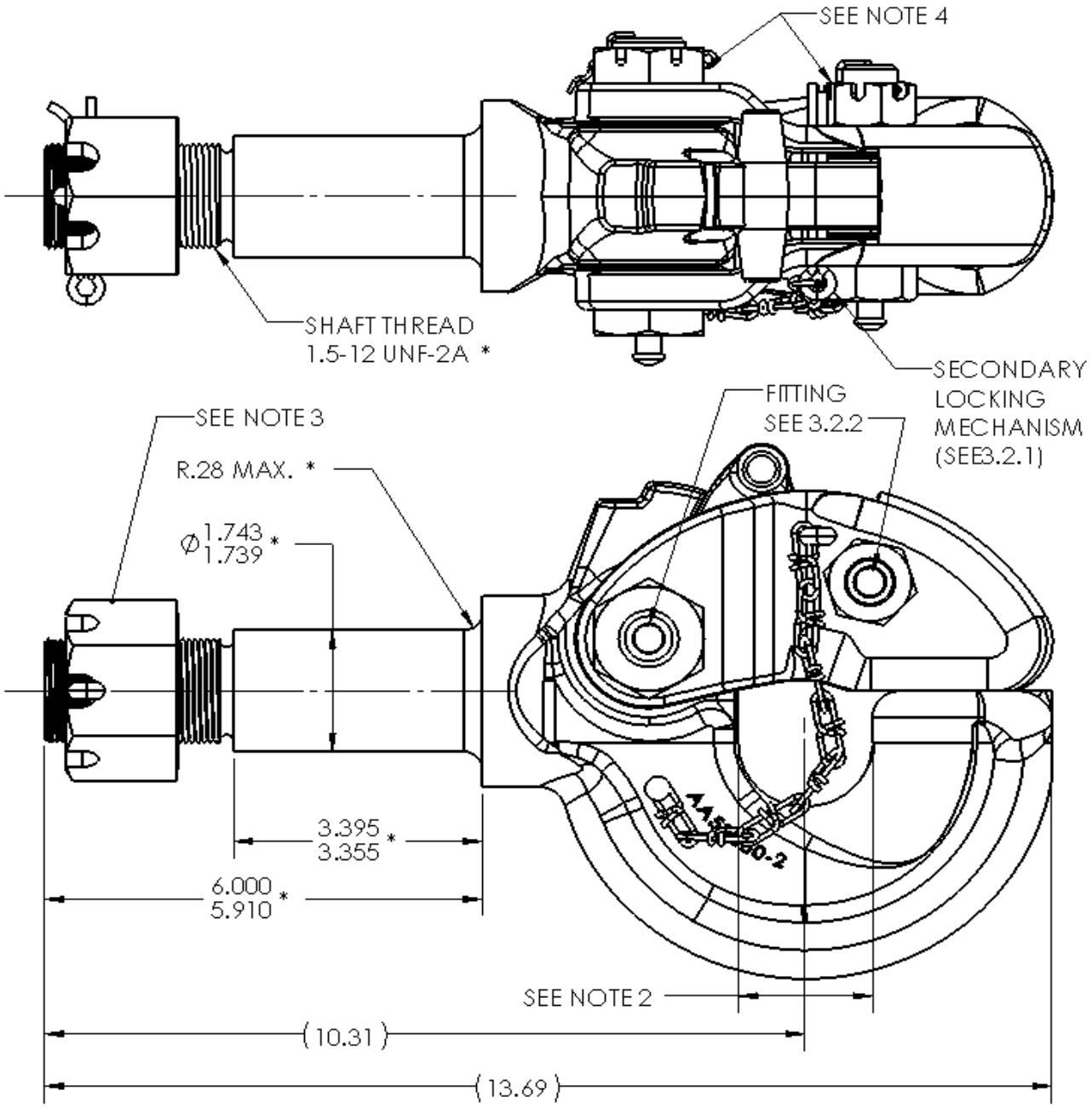
PIN		Former MS Part No.
May contain Cad & Cr6+	Cad & Cr6+ Free	MS51335-2
AA52550-4	AA52550-4E	

NOTES:

1. Dimensions are in inches. *These dimensions are required; all other dimensions are for engineering reference only.
2. The pintle throat shall be sized such that it will accept a 3 in. ID. X 1.66 dia. cross section lunette per STANAG 4101 (see paragraph 7.7).
3. Positive or prevailing locking device required to prevent loosening during service.
4. Latch mounting hardware (note 3) to comply with paragraphs 3.6 and 3.6.1.

FIGURE 2. Type I, class 2, pintle assembly.

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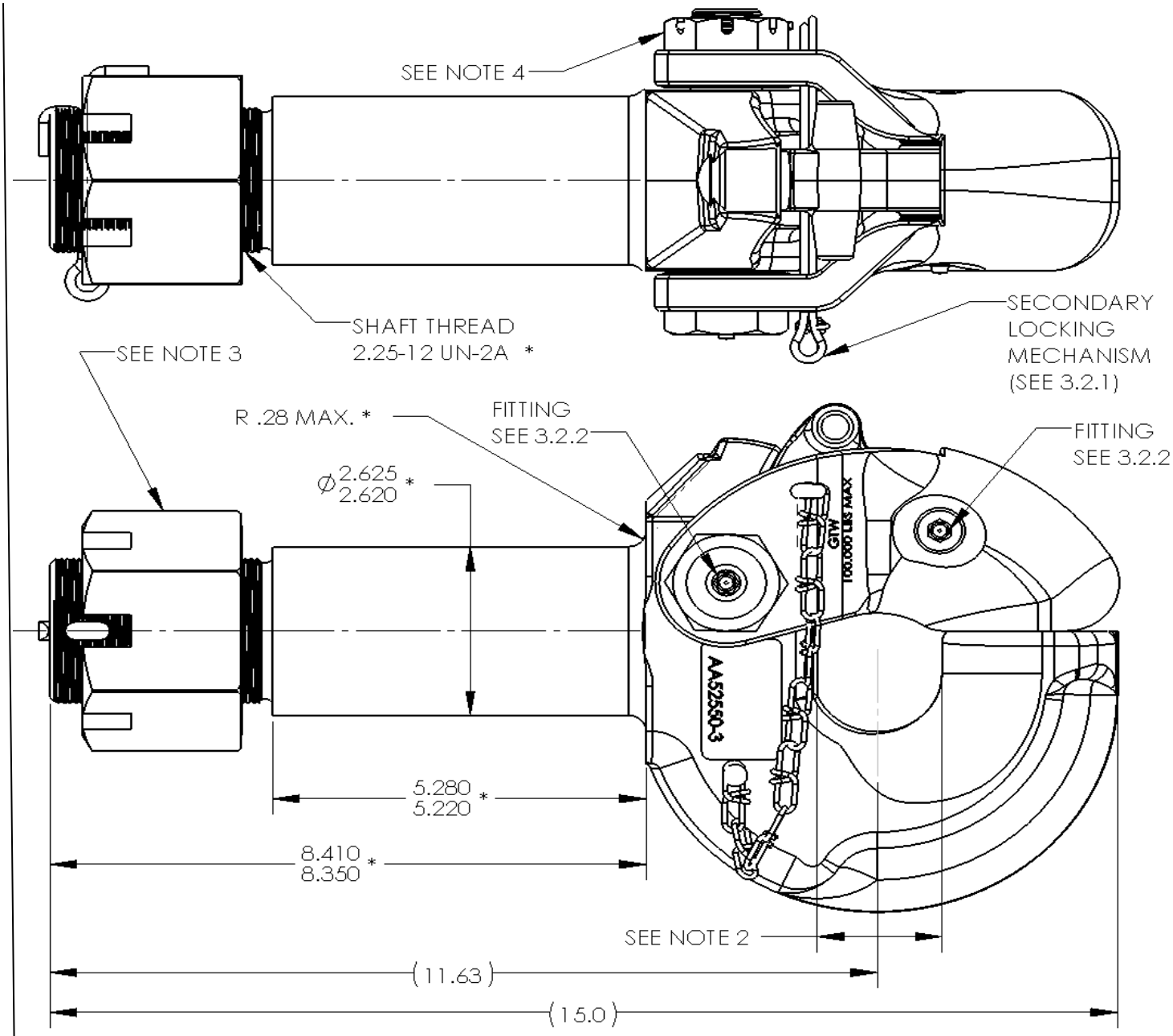
PIN		Former MS Part No.
May contain Cad & Cr6+	Cad & Cr6+ Free	MS51118-1
AA52550-2	AA52550-2E	

NOTES:

1. Dimensions are in inches. *These dimensions are required; all other dimensions are for engineering reference only.
2. The pintle throat shall be sized such that it will accept a 3 in. ID. X 1.66 dia. cross section lunette per STANAG 4101 (see paragraph 7.7).
3. Castellated nut, cotter pin, and shaft thread configuration are required as indicated.
4. Positive or prevailing locking device required to prevent loosening during service
5. Castellated nut and latch mounting hardware (note 3 & 4) to comply with paragraphs 3.6 and 3.6.1.

FIGURE 3. Type II pintle assembly.

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PIN		Former MS Part No.
May contain Cad & Cr6+	Cad & Cr6+ Free	MS51117-1
AA52550-3	AA52550-3E	

NOTES:

1. Dimensions shown are in inches, *These dimensions are required; all other dimensions are for engineering reference only.
2. The pintle throat shall be sized such that it will accept a 3 in. ID. X 1.66 dia. cross section lunette per STANAG 4101 (see paragraph 7.7).
3. Castellated nut, cotter pin, and shaft thread configuration are required as indicated.
4. Positive or prevailing locking device required to prevent loosening during service
5. Castellated nut and latch mounting hardware (note 3 & 4) to comply with paragraphs 3.6 and 3.6.1.

FIGURE 4. Type III pintle assembly.

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6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see paragraph 7.2).

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1. Addresses for obtaining copies of referenced documents.

7.1.1 Government documents. Copies of paragraph 23.403 "Policy on Use of Recovered Materials and Biobased Products" of the Federal Acquisition Regulation (FAR) are available from www.acquisition.gov. Copies of MIL-DTL-53072 "Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection" are available from https://assist.dla.mil/.

7.1.2 Non-Government documents. Copies of ASTM B117 "Standard Practice for Operating Salt Spray (Fog) Apparatus" are available from www.astm.org. Copies of SAE J534 "Lubrication Fittings" and SAE J847 "Trailer Tow Bar Eye and Pintle Hook/Coupler Performance" are available from www.sae.org. IFI standards are available at www.indfast.org.

7.1.3 International documents. Copies of STANAG 4101 "Towing Attachments" can be requested from the NATO Standardization Office website at https://nso.nato.int.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Type, PIN, and quantity of pintle assembly required.
d. Cleaning, surface treatment, type of paint, and color required.
e. Selection of applicable level and packaging requirements.

7.3 Part Identification Number (PIN). The PINs to be used for pintle assemblies acquired by this CID are created as follows:

AA52550 - 1 E

- E: Finish Code:
  - No code = Hexavalent chromium and cadmium content uncontrolled
  - E code = Free of hexavalent chromium and cadmium
- 1: 1 = Type I class 1; 2 = Type II; 3 = Type III; 4 = Type I class 2
- AA52550: CID designator

Example part number: AA52550-1E = Towing pintle assembly, 18,000 lb maximum gross trailer weight, swivel design, free of hexavalent chromium and cadmium.

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7.4 Pintle assembly (type I, class 2). A pintle assembly which uses a spring-loaded pin or a snapper pin as a secondary locking mechanism is recommended for use as type I, class 2, pintle assembly described in this CID because of unlatching problems associated with cotter pin type locking devices encountered in the field.

7.5 Supersession. This CID supersedes MS51335F, dated 29 September 1981; MS51118F, dated 30 May 1972; and MS51117E, dated 21 August 1972.

7.6 Cross-reference. Pintle assemblies conforming to this CID are similar to pintle assemblies conforming to MS51335F, MS51118F, and MS51117E; however they are not direct substitutes. It is permissible for pintle assemblies conforming to an AA52550-*E PIN (see paragraph 7.3) to be used in place of pintle assemblies conforming to the corresponding AA52550-* PIN (see paragraph 7.3).

7.7 International interest. Certain provisions of this CID (figures 1, 2, 3, and 4) are the subject of international standardization agreement (STANAG 4101). When amendment, revision, or cancellation of this CID is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

7.8 Changes from previous issue. The margins of this CID are marked with vertical lines to indicate where changes from the previous issues were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS:

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Navy - YD
Air Force - 184

Review Activities:
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Air Force - 84
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Army - AT

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